

Vulnerability analysis of water systems to power interruptions in critical infrastructures

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ABSTRACT

The operation of the water distribution system is highly dependent on the power supply. Previous studies that focused on static cascading failures in interdependent water and power distribution systems fail to account for their dynamic response to disturbances. To address this gap, this study develops a cascading failure analysis model to simulate their dynamic response to disturbances. Vulnerability assessment indicators are also proposed based on Water Service Capability (SCW) and Power Service Capability (SCP) evolution curves under disturbance scenarios to quantify ripple effects of power interruptions on water distribution systems. Using a 15-node testing water distribution system with a pumping station power supply system as an example. Generating storm scenarios for time-series simulation using Monte Carlo sampling. Results show that SCW exhibits short-term vulnerability to power interruptions but maintains robust buffering and recovery capabilities over the medium to long term. SCW loss duration is on average 4.5 times longer than SCP. Storage tanks delay water supply loss to demand nodes about 4 h longer than the pump failure time. Vulnerability of water distribution systems increases with longer pump failure durations. The proposed model can provide decision support for reducing cascading vulnerability of interconnected power and water networks to cascading failures by allowing alternative designs and modifications to be simulated.

1. Introduction

Critical infrastructures are needed by human societies for their basic needs, including transportation, water supply, power supply, and communications. These infrastructures are usually highly interdependent [1]. For example, water distribution systems are highly reliant on power transmission for their normal operation [2]. Power systems also depend on a steady supply of water for cooling and steam generation on thermal plants, and as the working fluid in hydroelectric plants [3,4]. Natural gas distribution requires electric motor-driven gas compressors to maintain acceptable gas pressures for normal gas supply [5]. The widely available water-electricity interconnected networks [3], water-electricity-gas interconnected networks [5,6], electricity-communication networks [7], and water supply-transport networks [8] provide basic needs for

society to function. However, mutually interdependent infrastructure is vulnerable to cascading failure effects; vulnerability also increases with system complexity, particularly when feedback loops are present [9,10]. For example, Hurricane Sandy directly caused the collapse of the power and transportation infrastructure in the United States in 2012; these disruptions then affected the infrastructure supplying water, communication, and healthcare services [11]. Similar disruptive events include the adverse weather conditions in southern China in 2008 [12] and Hurricane Maria in Puerto Rico in 2017 [13]. In addition to natural disasters, manmade disruptions such as the Suez Canal blockage [14] and malicious attacks [15] can also threaten infrastructure systems. Thus, studying cascading failure mechanisms among interconnected infrastructures is important for disaster mitigation and prevention [16].

Power and water distribution systems as typically interdependent

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networks, have attracted much attention in recent years. Their stability and reliability are essential to ensure that basic human needs are met in any given region. Water distribution system is mainly composed of elements such as water source, water treatment plant, pipelines, joints, storage tanks, reservoirs, pumps, and valves [17]. Our previous work simulated the disturbance propagation trajectory among components in water networks using the dynamic inoperability input-output model [18]. Extracted water from the source depends on power to support the pumps to deliver to the users, and the same power system depends on water to cool the generators or to produce energy [19]. Safe operation of water infrastructure, particularly for pumping and treatment, highly depends on the availability of electricity [20,21]. 7–8% of total electricity generation is consumed for drinking water production and distribution systems in China [22].

Because power and water systems are interdependent, managing them separately can lead to ineffective results; instead, they need to be analyzed as integrated systems [23,24]. There is a body of research literature focusing on the analysis and design of interdependent networks to achieve better performance. For example, Alhazmi et al. [25] proposed a cross-sectoral water and power system optimization framework that achieves significant cost savings. Oikonomou and Parvania [26] developed a network constrained unit combination model that ensures energy flexibility and stable conveyance capacity for water and power distribution systems. Mohebbi et al. [8] used a coalitional game approach to solve the problem of decentralized resource classification for interdependent water-road networks. Zuloaga and Vittal [27] proposed a resilience metric for coordinating the joint operation of water and power distribution systems. However, due to the continuous attacks on interdependent networks by natural disasters and man-made failures, once the power distribution system is interrupted, it inevitably affects the normal operation of water distribution systems. For example, the widespread power outages in the northeastern United States and parts of Canada in 2003 resulted in secondary failure of water systems [28]. Therefore, analyzing the vulnerability or cascading failures of interdependent power and water distribution networks is essential for better coordinating their operations.

Previous studies on cascading vulnerability focused mainly on two perspectives: static and dynamic. Static assessments primarily relied on network topology theory, utilizing structural metrics such as connectivity [29], betweenness centrality [30], and closeness centrality [31, 32] to evaluate cascade vulnerability at nodes within water distribution networks. Functional coupling strength has been employed to assess the likelihood of fault propagation between components, thereby evaluating cascade vulnerability in power and water systems [33]. However, it overlooks the dynamic evolution of water node disturbances. To address this, researchers focused on simulating the performance curve of the system under disaster disturbances [34]. On this basis, vulnerability assessment metrics were developed, accounting for temporal variability, based on the probability of exceeding critical tolerance thresholds [35] and performance level deviation thresholds [36]. Furthermore, it characterizes the vulnerability of water distribution systems using the demand fulfillment rate [37]. However, it does not consider the vulnerability of integrated water and power distribution systems. In addition, existing studies mostly focus on failure cascade failure modes [35,37], but water distribution networks more consistently fit non-failure cascade modes. Because it is often equipped with alternate water sources or other buffers, water flows are gradually lost, which clearly limits the insights that can be drawn from disturbance propagation and recovery measures changes.

To address above research gaps, this study aims to reveal non-failure cascading disturbance propagation mechanism of power interdependent to water networks based on full-process curves for disaster response, and establish a full-process vulnerability assessment methodology. Focus on addressing the following questions: How does power system disruption propagate through coupling relationships to the water distribution system over time under extreme disaster scenarios? What are the evolution

patterns? Which characteristics of power outages—such as duration and intensity—have the most significant impact on the water distribution system? Therefore, the main research content is as follows. Firstly, an analytical model of power interruption cascade water shortage is developed to describe the network of physical connections between water and power distribution systems. Proposing a power outage-cascading water shortage analysis model to simulate the full process of power interruption and how to affect the normal operation of water distribution system. Secondly, using Monte Carlo sampling to randomly generate storm scenarios, with settings for random line faults, vulnerability coefficients, and fault repair times. Finally, building upon system performance curves, we establish comprehensive vulnerability assessment metrics covering the entire process (resistance, degradation, recovery) to evaluate the cascading effect. By identifying critical influencing factors and risk characteristics, we provide decision-making support for risk management and resilience enhancement.

The paper is organized as follows. Section 2 presents the power interruption cascading water shortage model, including the introduction of failure propagation mechanisms, mathematical models, and vulnerability indicators. Section 3 introduces the basic information of the selected case for this study and sets up the corresponding power interruption disturbance scenarios. Section 4 gives the simulation results and analysis. Section 5 discusses the research importance, advantages, and limitations. Section 5 provides the conclusion and discusses future research directions.

2. Vulnerability analysis for water and power systems

2.1. Failure propagation mechanism

The propagation mechanism of water supply interruptions resulting from power interruptions is shown in Fig. 1. When an interruption occurs on one of the lines in the power distribution system, the load point of the line is de-energized. If the interrupted line connected the pumps of the water distribution system, followed by loss of energy for the water pumps. If the pump plays the delivery role in the water distribution system, which finally result in the inability to deliver the water to the given water demand nodes. If the water demand node does have a storage tank (back-up water), the node suffers a water shortage. Therefore, the actual water shortage at the water demand node is determined by pump status and water storage tank capacity [38]. In this paper, the water shortage in water demand nodes of the water distribution system caused by the breakdown of the power distribution system is defined as a cascading effect.

2.2. Cascading model

Based on the failure propagation mechanism introduced previously, a mathematical model of the cascading effect is developed. Eq. (1) and (2) indicate the water flow balance and constraints at the water demand node. Actual water supply $q_i(t)$ at node i in time t is determined by the difference between the total inflow and the total outflow for node i . $Q_{ji}(t)$ is the water flow from node j to node i by the pump, and $Q_{ik}(t)$ is the water flow from node i to node k by the pump. $D_n(t)$ represents water demand at node i :

$$\sum_{j \in \text{in}(i)} Q_{ji}(t) - \sum_{k \in \text{out}(i)} Q_{ik}(t) = q_i(t) \quad (1)$$

$$0 \leq q_i(t) \leq D_i(t) \quad (2)$$

Define the node water supply rate in time t as $\gamma_n(t)$:

$$\gamma_n(t) = \begin{cases} \frac{q_n(t)}{D_n(t)} & D_n(t) > 0 \\ 1 & D_n(t) = 0 \end{cases} \quad (3)$$

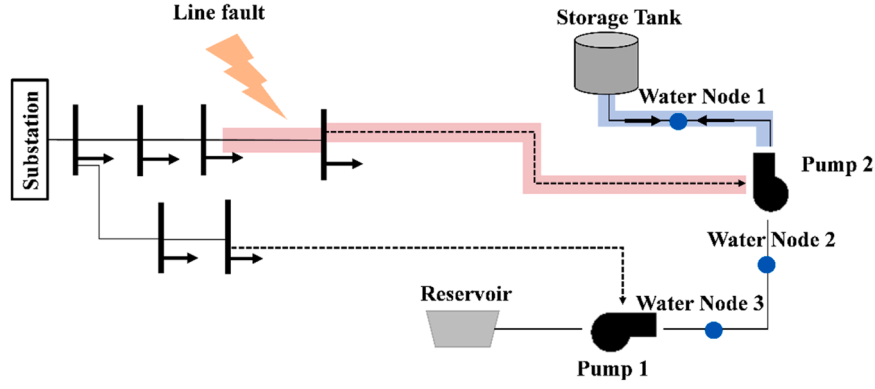


Fig. 1. Fault propagation from power to water distribution system.

Eq. (4) indicates that $\gamma_p(t)$ is whether pump p is powered at time t , where the connection to the line is determined by graph connectivity. Eq. (5) expresses that the pump flow $Q_p(t)$ is constrained by the pump's maximum flow rate and its operational status:

$$\gamma_p(t) = \begin{cases} 1 & \text{bus}(p) \in \text{Reach}(t) \\ 0 & \text{bus}(p) \notin \text{Reach}(t) \end{cases} \quad (4)$$

$$Q_p(t) = \begin{cases} \min(Q_p^{\max}(t), D_p(t)) & \gamma_p(t) = 1 \\ 0 & \gamma_p(t) = 0 \end{cases} \quad (5)$$

Eq. (6) indicates that pump power $P_p(t)$ is determined by the pump flow rate $Q_p(t)$ and pump head $H_p(t)$. Pump head is the pressure head difference between outlet and inlet (Eq. (7)), which is determined by pipeline resistance losses (using Hazen-Williams' formula and the pump curve (Eqs. (8)-(9)):

$$P_p(t) = \frac{\rho g Q_p(t) \cdot H_p(t)}{\eta} \quad (6)$$

$$H_p(t) = H_{p,\text{out}}(t) - H_{p,\text{in}}(t) \quad (7)$$

$$H_{p,\text{out}}(t) - H_{p,\text{in}}(t) + h_f(t) = a + b Q_p(t)^c \quad (8)$$

$$h_f(t) = 10.67 \cdot \frac{Q_p^{1.852}(t)}{C^{1.852} \cdot d^{4.87}} \cdot L \quad (9)$$

$$P_p(t) \leq P_p^{\max} \quad (10)$$

Where ρ is water density, 1000 kg/m³; g is gravity acceleration, 9.81 m/s²; and η is the pump efficiency. It measures the conversion efficiency of the pump in transforming input mechanical energy into fluid hydraulic energy. Due to energy losses during transmission, this efficiency is typically less than 1. a is the shut-off head. b, c are the fitting parameters for the pump. $H_{p,\text{in}}(t)$ is the head at the pump inlet node. $H_{p,\text{out}}(t)$ is the head at the pump outlet node. h_f indicates the head loss along the pipeline. C is the roughness coefficient, 80. d is the inner diameter of the pipe, L is the length of the pipe.

When pumps cannot meet water demand or supply is insufficient, water tanks serve a critical role in filling gaps or alleviating peak demand pressure on pumps. Eqs. (11) to (13) represent the operational water balance and constraints for water tanks:

$$V_T(t) = V_T(t-1) + Q_T^{\text{in}}(t) - Q_T^{\text{out}}(t) \quad (11)$$

$$V_{T,\min} \leq V_T(t) \leq V_{T,\max} \quad (12)$$

$$|Q_T(t)| \leq Q_{T,\max} \quad (13)$$

Where $V_T(t)$ is the volume of the water tank T at time t . $Q_T^{\text{in}}(t)$, $Q_T^{\text{out}}(t)$ are the inflow and outflow rates of tank T at time t , determined by the pump

flow rate and water node demand. $V_{T,\min}$, $V_{T,\max}$ are the minimum and maximum volumes of the water tank T . $Q_{T,\max}$ is the maximum charging and discharging rate of the water tank T .

2.3. Vulnerability analysis

Water Service Capability (SCW) is calculated as the weighted average based on actual supply and demand at each water demand node (Eq. (14)). When SCW equals 1, it indicates that the water supply to the water demand nodes meets the standard. Lower values indicate more severe water supply shortages. Similarly, Power Service Capability (SCP) is calculated as the weighted average based on the maximum flow rate of each pump (Eq. (15)). If power outages occur on lines connected to the pump, it poses greater threat to the water distribution system. This metric indicates the electricity reliability rate of the potential water supply capacity:

$$SCW(t) = \frac{\sum_{n \in N} q_n(t)}{\sum_{n \in N} D_n(t)} \quad (14)$$

$$SCP(t) = \frac{\sum_{p \in P} Q_p(t)}{\sum_{p \in P} Q_p^{\max}(t)} \quad (15)$$

To assess the primary contribution of each pump or water demand node to SCP and SCW, the marginal contribution of each pump and water demand node to SCP and SCW is defined as follows:

$$\text{contrib}_p(t) = \gamma_p(t) \cdot w_p(t) \quad (16)$$

$$\text{contrib}_n(t) = \gamma_n(t) \cdot w_n(t) \quad (17)$$

$$w_p = \frac{Q_p(t)}{\sum_{p \in P} Q_p^{\max}(t)} \quad (18)$$

$$w_n = \frac{q_n(t)}{\sum_{n \in N} D_n(t)} \quad (19)$$

Subsequently, we use loss attribution to measure which pump and water demand nodes contribute to greater SCP and SCW losses:

$$\text{Loss}_p(t) = (1 - \gamma_p(t)) \cdot w_p(t) \quad (20)$$

$$\text{Loss}_n(t) = (1 - \gamma_n(t)) \cdot w_n(t) \quad (21)$$

According to the failure propagation mechanism, the typical evolution curves of $SCW(t)$ and $SCP(t)$ under interruption scenarios are shown in Fig. 2 (a) and (b). Both $SCW(t)$ and $SCP(t)$ experienced four evolution stages: pre-interruption stage, degradation stage, recovery stage, and post-interruption stage. Initial faults occur on the distribution system lines at t_0^p . $t_0^p \sim t_m^p$ is the line fault duration, described as the SCP degradation phase, where t_m^p is the minimum SCP moment. The beginning of $SCW(t)$ interruption t_0^w depends on whether the line to be

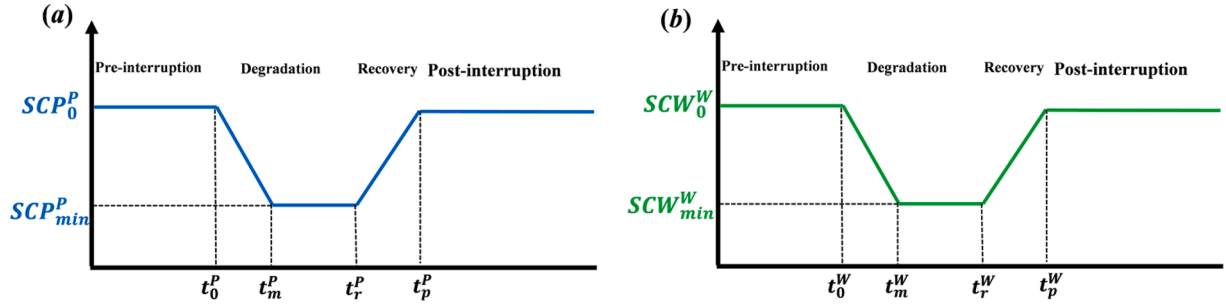


Fig. 2. General evolution curves of SCP (a) and SCW (b).

interrupted is connected to the pump and the water available in the storage tank. $t_0^W \sim t_r^W$ is the water supply interruption duration, described as the SCW degradation phase, where t_m^W is the minimum SCW(t) moment. Depending on the intensity of the interruption scenario, there exist $SCW_{min}(t)$ and $SCP_{min}(t)$. $t_r^P \sim t_p^P$ is the SCP interruption repair or recovery phase, where the time taken is determined by the maturity of the repair technology. $t_r^W \sim t_p^W$ is the SCW recovery phase.

Three vulnerability assessment indicators are developed based on SCW and SCP evolution curves to quantify the cascade effects of power distribution system interruptions on water distribution systems. RD and RQ denote the power-water coupling characteristics during the degradation and recovery phases, respectively. RA represents the overall impact level of the power distribution system on the water distribution system.

RD is the degradation dominance index, measuring the amplification of SCW relative to SCP during the system degradation phase. It reflects the sensitivity of water distribution system performance decline following power outages. Its value ranges from 0 to 1. At 0.5, both power and water sides contribute equally to degradation. Higher values indicate that SCW loss areas are more severely impacted than SCP losses during degradation phases, signifying greater vulnerability/sensitivity of the water system and weaker tank buffering capacity:

$$RD = \frac{A_{SCW}^{deg}}{A_{SCW}^{deg} + A_{SCP}^{deg} + \varepsilon} \quad (22)$$

$$A_{SCW}^{deg} = \int_{t_0^W}^{t_m^W} (1 - SCW(t)) dt \quad (23)$$

$$A_{SCP}^{deg} = \int_{t_0^P}^{t_m^P} (1 - SCP(t)) dt \quad (24)$$

RQ is the recovery dominance index, measuring the prevalence of lagged effects during the system recovery phase. It reflects the ability of SCW to return to a steady state after power line restoration. Its value ranges from 0 to 1, where 0 indicates SCW recovers almost simultaneously with SCP. Higher values indicate a larger recovery gap area relative to SCP, signifying slower or lagged recovery:

$$RQ = \frac{A_{SCW}^{rec}}{A_{SCW}^{rec} + A_{SCP}^{rec} + \varepsilon} \quad (25)$$

$$A_{SCP}^{rec} = \int_{t_r^P}^{t_p^P} (1 - SCP(t)) dt \quad (26)$$

$$A_{SCW}^{rec} = \int_{t_r^W}^{t_p^W} (1 - SCW(t)) dt \quad (27)$$

RA is the system robustness index, measuring the ability of SCW to maintain service relative to SCP fluctuations throughout the entire disaster cycle. If the value is less than 1, it means the water distribution system is more degraded compared to the power distribution system,

while a value greater than 1 means that water demand for meeting water distribution system operations is less dependent on the power distribution system.

$$RA = \frac{\int_0^T (1 - SCW(t)) dt}{\int_0^T (1 - SCP(t)) dt} \quad (28)$$

3. Case study

3.1. Base information

The case study we selected is the IEEE 33-bus power distribution and 15-node water distribution test system with desalination plant, which was developed by Utah Smart Energy Lab [39]. This case has been widely applied to research related to water and power networks. Such as network optimal design to reduce cost [24], decrease pollution [40], and increase the observability and controllability for power and water distribution system operations [41]. The system network is shown in Fig. 3. The power distribution system includes 33 buses (b1, b2, ..., b33) and 32 Lines (L1, L2, ..., L32). Table A1 contains the power distribution lines data, used to determine line connectivity. b33, b10, and b22 directly supply power to Pumps 1, 2, and 3. These pumps are respectively located on pipelines J1, J4, and J7. Table A2 lists the pump performance characteristic function data, which is used for calculating pump head, power, and flow rate balance. Water distribution system includes 15 water nodes (N1, N2, ..., N15) and 14 pipes (J1, J2, ..., J14). Among these are two water storage tanks (N10, N13) and three typical water

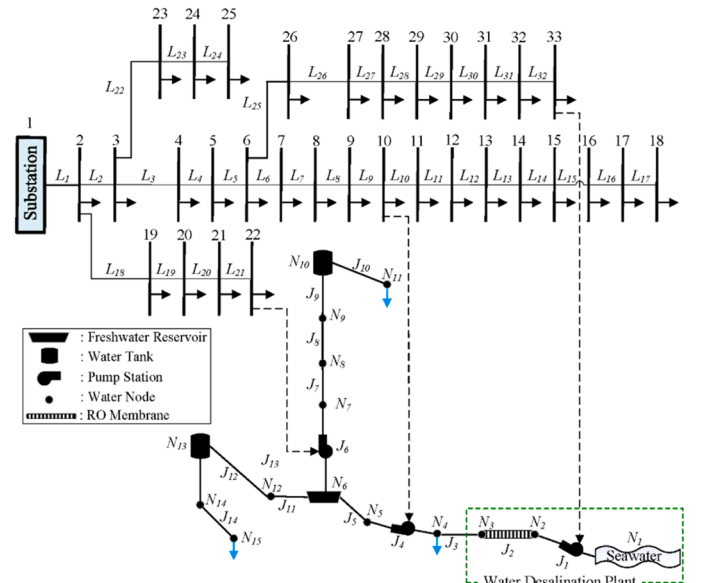


Fig. 3. Interconnected 33-bus power test system and 15-node water distribution system.

demand nodes (N4: residential, N11: commercial; N15: industrial). The elevation and pressure of each water node are listed in Table A3. Tables A4 and A5, respectively, list the original water pipes data and water storage tank characteristic data. These are used for model calculations.

Taking one day as a period, the power demand curve for load lines of the power distribution system and water demand curve for nodes of water distribution system are shown in Fig. 4. Water demand fluctuates within a day. Industrial node N15 experiences substantial water demand, peaking at 2076.8 m³/h (12:00). Commercial node N11 and Residential node N4 exhibit daytime consumption peaks with lower nighttime usage, reaching maximum demands of 1629.4 m³/h (12:00) and 688.45 m³/h (8:00), respectively. The load power demand for line b22 is approximately two times higher than for lines b10 and b33. b22, b10, and b33 all experience higher power demand during the morning (8:00–12:00) and evening (18:00–22:00) periods. Peak power demand reaches 60 kW, 90 kW, and 90 kW (at 19:00), respectively. The source data comes from Utah Smart Energy Lab [39].

3.2. Scenario setting

To quantify the cascading effects of power interruptions on the water distribution system. We simulate the impact of extreme storms on the line reliability of power distribution systems using Monte Carlo methods. The storm process is modeled by a single-peak Gaussian wind speed curve, where the wind speed reaches its peak value $\omega_{\max}$ at time μ , with temporal dispersion controlled by σ :

$$\omega(t) = \omega_{\max} \exp\left(-\frac{1}{2} \left(\frac{t-\mu}{\sigma}\right)^2\right), \quad t = 1, 2, \dots, T \quad (29)$$

For each line, assign a vulnerability coefficient ϕ_l , where larger coefficients indicate greater susceptibility to failure during storms. These coefficients are randomly sampled from the interval [0.05, 0.3]:

$$\phi_l \sim \text{Uniform}(0.05, 0.3), \quad \forall l \in \mathcal{L} \quad (30)$$

Set the probability of line failure at each time t as the linear function of wind speed, where $p_l(t)$ increases linearly with wind speed $\omega(t)$, and impose an upper bound of 0.95 to prevent excessive probability values causing numerical instability or total failure:

$$p_l(t) = \min\left(0.95, \phi_l \cdot \frac{\omega(t)}{\omega_{\max}}\right), \quad \forall l \in \mathcal{L}, t = 1, 2, \dots, T \quad (31)$$

When a line failure occurs, the repair time τ_l follows the normal distribution, characterizing the variability in repair duration caused by uncertainties in repair resources and environmental conditions. Minimum constraints and integerization are applied:

$$\tau_l \sim \max(1, \text{Round}(\mathcal{N}(\mu_r, \sigma_r))) \quad (32)$$

Define the line outage status matrix (Eq. (33)). Using the Monte Carlo method, generate outage sequences for each line and each hour. For each line in time t , sample a random variable:

$$x_l(t) = \begin{cases} 1, & \text{Line Outage/Fault/Repair} \\ 0, & \text{Normal operation} \end{cases} \quad (33)$$

$$U_{l,t} \sim \text{Uniform}(0, 1) \quad (34)$$

$$\text{If } U_{l,t} \leq p_l(t), x_l(k) = 1, \quad k = t, t+1, \dots, \min(T, t+\tau_l) - 1 \quad (35)$$

Within the continuous outage interval, the system will not trigger new fault determinations until the line is restored to operation. Finally, the line outage matrix is obtained:

$$X = [x_l(t)] \in \{0, 1\}^{\mathcal{L} \times T} \quad (36)$$

In this study, the maximum wind speed was set to 35, the storm peak time to 12, the standard deviation to 1, the mean repair time to 2, and the standard deviation of repair time to 1. Results show that wind speeds exhibited fluctuating changes (Fig. 5), with high-wind periods concentrated between 10:00 and 16:00.

4. Results

4.1. Power distribution system fault analysis

Line failure frequencies exhibit significant heterogeneity among scenarios (Fig. 6(a)). Trunk lines such as Line 5–7, Line 11, and Line 14 demonstrate consistently high failure frequencies exceeding 0.9 in all scenarios. In contrast, lines including Line 2–4 and Line 26 exhibit relatively low failure frequencies around 0.8. High-frequency failures occurred simultaneously on numerous lines under Scenario 31, such as Lines 11–18 and Lines 26–32, while Scenario 14 only showed short-term failures on a few lines. The cumulative power outage hours varied significantly across different scenarios for each pump in the system (Fig. 6(b)). P1 had the longest cumulative failure duration at 231 h, with the highest single-scenario failure lasting up to 9 h (Scenario 41). P2

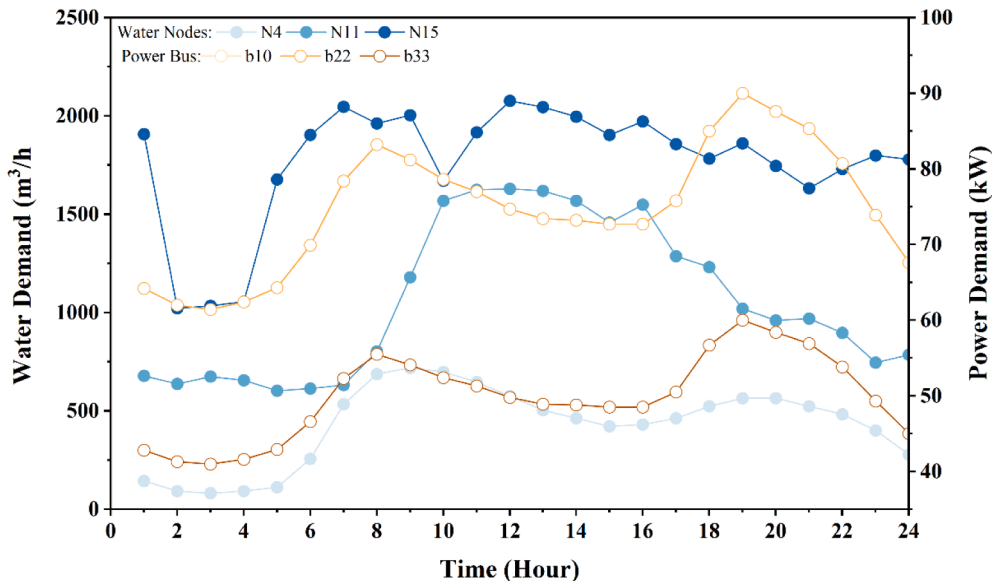


Fig. 4. Power and water demand curve for one day.

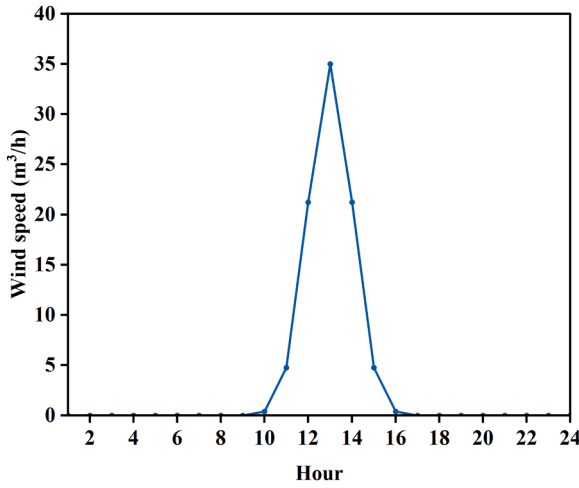


Fig. 5. Wind speed curves.

follows with 203 h, and the maximum single-scenario failure duration reaches 7 h (Scenarios 9 and 21). While P3 had the shortest cumulative failure duration, only 139 h, with an average of approximately 2.5 h per scenario. High-frequency failure lines correspond to long outage durations for pumps. Since L32 directly supplies power to P1 and exhibits longer single-fault durations, P1 experiences the most severe failures. The power supply line for P2 (L10) is governed by the main trunk lines L1–L9, making its vulnerability equally significant. Line 22 supplying P3 exhibits relative independence, influenced only by L1 and branch lines L18–L21. This confirms the lower outage duration observed for P3.

Based on the developed model for an interdependent water and power distribution system, modelling obtains the evolution curves of SCP for all scenarios as shown in Fig. 7. SCP evolution curves exhibit minor U-shaped patterns (Fig. 7(a)). Consistent with wind speed disturbance curves, nighttime represents a stable period. Performance declines between 10:00 and 13:00 on average, recovers between 14:00 and 18:00, and reaches its lowest value of 0.154 at 13:00. Scenario 41 exhibits the earliest SCP decline onset (10:00), while Scenario 43 first reaches minimum service capacity (11:00). During full-load operation, P3 serves as the core support, contributing approximately 42% of SCP (Fig. 7(b)), followed by P2 and P1, contributing about 35% and 23%, respectively. During the SCP degradation period, P3 continues to contribute significantly, accounting for 42.03%–86.32%, with the highest proportion occurring at 14:00. Notably, P2 contributed the most to SCP loss (Fig. 7(c)), ranging from 37.45%–60.06% with peak occurrence at 10:00, indicating significant negative impact from P2 flow loss

on the system. Loss contributions from P1 and P3 fluctuated considerably, ranging from 27.11%–57.08% and 0%–35.44%, respectively. P2 is the primary driver of SCP degradation (Fig. 7(d)), accounting for 54% of the dominant loss scenarios. This is attributed to the substantial water demand at node N15 supplied by P2. P3 follows as the next significant contributor, while P1 exerts a relatively minor impact on overall system instability.

4.2. Water distribution system fault analysis

Various pump flow rates exhibit distinct time-varying characteristics within a day (Fig. 8(a, b, c)), with average flow rate of P2 highest at 1429 m³/h, primarily due to the high-water demand of N15 it supplies. P3 follows at 877 m³/h, while P1 is 332.76 m³/h. Furthermore, the flow variation of P2 across different scenarios is substantial. Its 95% confidence interval exhibits extreme expansiveness between 12:00 and 16:00, even approaching near-zero in some scenarios. This indicates that P2 is most sensitive to disturbances in the power supply lines, consistent with the earlier finding that P2 dominates SCP losses. While the variation in P1 flow across different scenarios is relatively minor. Water demand nodes also exhibit distinct time-varying characteristics in their daily water supply volumes (Fig. 8(d, e, f)). The average daily losses at N4, N11, and N15 were 2278 m³/h, 789 m³/h, and 3265 m³/h, respectively. Notably, the loss at N15 corresponded to the highest flow loss from its supply pump P2. Actual water supply curves generally follow pump flow variations, but due to the buffer effect of the storage tank, the onset of water supply loss at demand nodes occurs significantly later than the loss in pump flow. For example, while loss in P2 flow began on average at 11:00, the average loss at supply node N15 started at 13:00. The storage tank's buffering effect against supply interruptions thus lasted for 4 h (Fig. 8(h)). Since N4 has no water tank to buffer it, its water supply fluctuations closely mirror the flow rate of P1. Tanks N10 and N13 effectively served as short-term buffers, both being heavily utilized to supply water to demand nodes between approximately 14:00 and 18:00 (Fig. 8(g, h)). N13 experienced an average volume decrease of approximately 4500 m³ between 14:00 and 16:00, while N10 decreased by about 3300 m³ during the 15:00–16:00 period. The wider confidence interval for N13, coupled with its frequent approaches to zero, indicates that N13 is depleted in most scenarios, with its buffer capacity insufficient to compensate for the flow loss gap in P2. Therefore, targeted reinforcement of P2 can significantly enhance the system's robustness.

SCW evolution curve exhibits distinct V-shaped patterns (Fig. 9(a)), with nighttime as the stable period. Between 11:00 and 15:00 on average, SCW experiences degradation, reaching its lowest point two hours later than SCP at 0.465. From 16:00–19:00, performance recovers. Consistent with SCP, SCW in Scenario 41 began experiencing

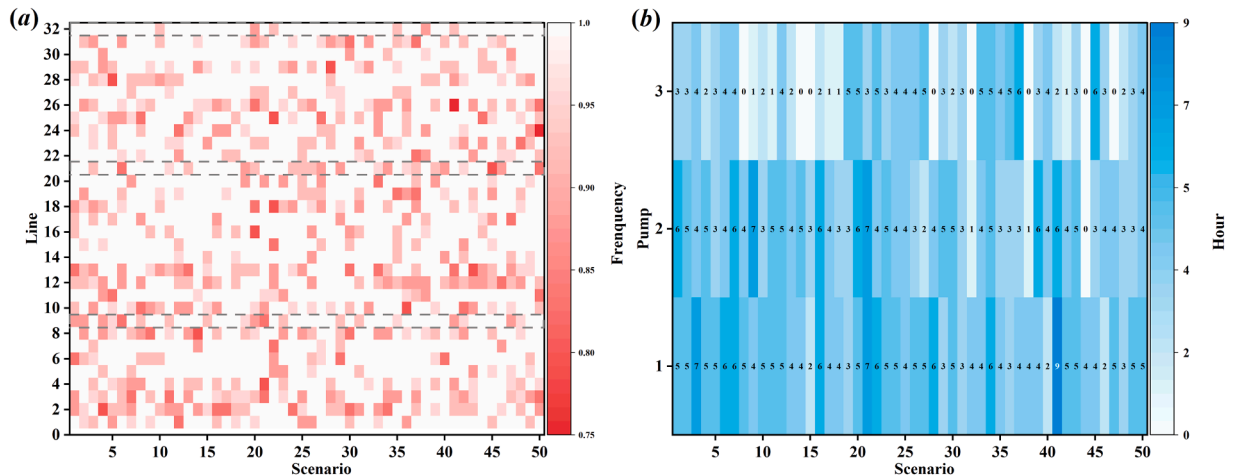


Fig. 6. Line failure frequency under all scenarios (a) (0–1,1 indicates 24-hour failure). (b) power outage duration for pumps P1, P2, P3.

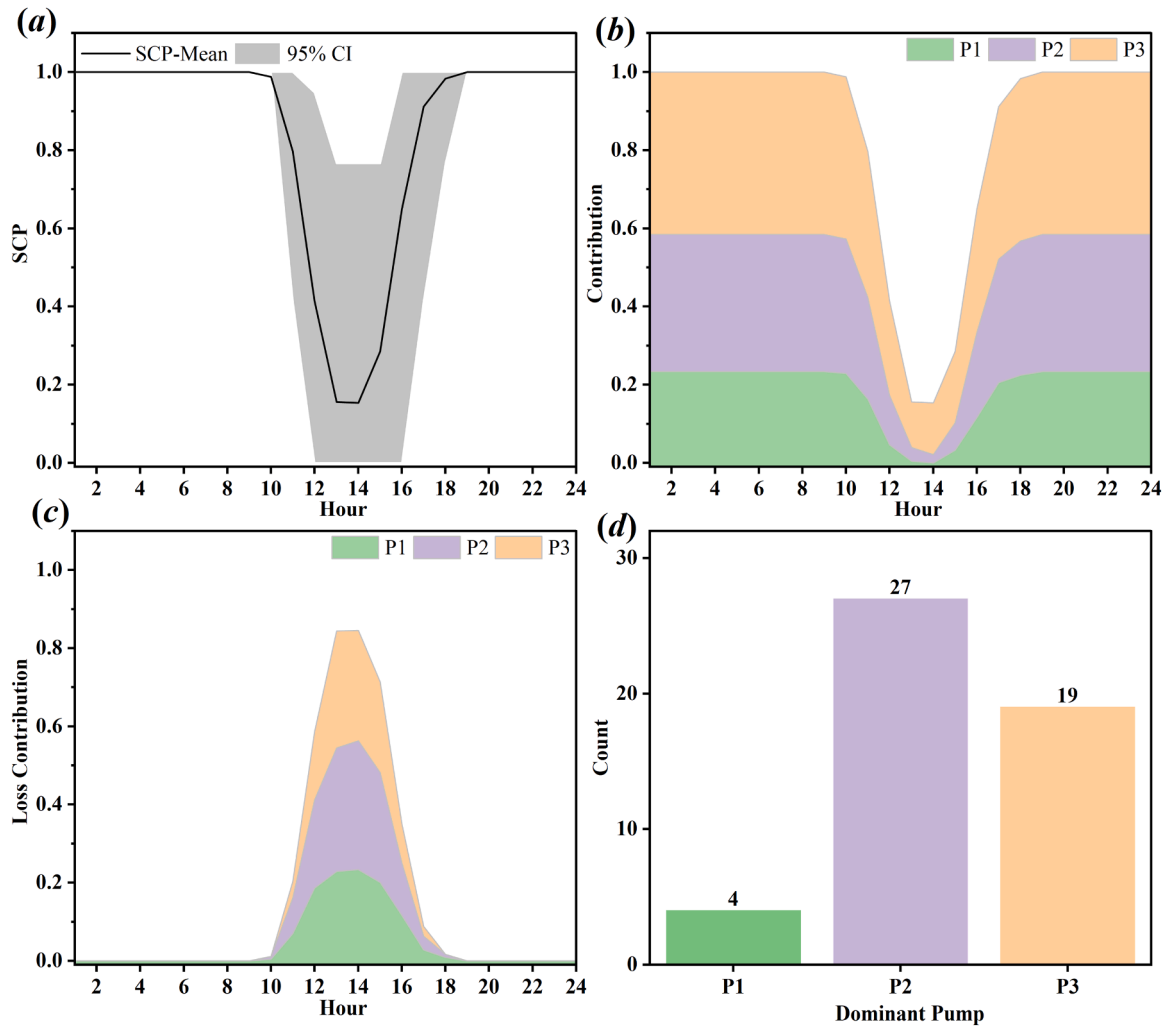


Fig. 7. SCP mean evolution curves under all scenarios (a); (b) contribution of each pump flow rate to SCP; (c) contribution of each pump flow rate to SCP loss; (d) number of scenarios where each pump dominates SCP loss.

degradation at 10:00. However, the narrow confidence interval and steep recovery curve indicate that the water tank enabled the water system to recover rapidly. When water supply is normal, N15 dominates contributions to SCW, averaging 50.51% (Fig. 9(b)). Due to its highest water demand, its contribution remains dominant even during SCW degradation periods, exceeding 42%, indicating its water deficit is promptly replenished. Next is N11, with an average contribution of 31.81%. At 16:00, its contribution even reached 53.11%. N4 made a relatively minor contribution to SCW, averaging approximately 10%. Notably, during the degradation period, N4 and N15 alternated as the primary contributors to SCW losses (Fig. 9(c)). Between 11:00 and 14:00, N4 contributed over 66% of SCW losses, while from 15:00–19:00, the contribution of N15 exceeded 60%, indicating a significant weakening of the buffering capacity of tank N13 for N15 during this period. Overall, N15 dominates SCW losses in the 60% scenario (Fig. 8(d)), far exceeding N4 (32%) and N11 (8%). This highlights the critical role of N15 in the degradation of water supply service capacity. Prioritizing the reliability of water supply at the N15 node is essential, including enhancing the supply capacity of P2 and N13.

4.3. Vulnerability assessment

Based on the SCP and SCW evolution curves, combined with proposed vulnerability assessment indicators, evaluate the distribution characteristics of RD, RQ, and RA for all scenarios (Fig. 10). RD mean

value of 0.55 indicates that the degradation phase is mildly dominated by the water distribution system, primarily due to the water tank creating a certain delay-steep-drop effect, resulting in a relatively larger degradation area for the SCW curve compared to the SCP. For scenarios where $RD > 0.5$, the SCP degradation period is relatively short, averaging approximately 0.78 h, while the SCW duration reaches as high as 3.48 h. However, for scenarios where $RD < 0.5$, the average degradation durations for SCP and SCW are comparable, at 1.65 h and 2.17 h, respectively. With an average RQ value as low as 0.15, the overall distribution indicates that in most scenarios, the system performance gap is primarily determined by SCP, while SCW does not constitute a significant recovery bottleneck. The average recovery duration for SCP is longer than that for SCW, at 8.68 h and 7.7 h, respectively, demonstrating that SCW recovers more rapidly. RA denotes the robustness distribution of the system, with a mean value of 1.1 and generally exceeding 1. This indicates that due to water storage tanks and pipeline network inertia, SCW can maintain a certain service level even when SCP declines at specific times, demonstrating the overutilization of available electricity.

Spearman correlation analysis indicates that RD exhibits a significant positive correlation with the interruption durations of P1 and P2 (Fig. 11), while the relationship with P3 is not significant. This suggests that RD is primarily governed by the duration of failure of the critical pumps P1 and P2. In Scenarios 3 and 21, the duration of P1 failure reached 7 h (12:00–18:00 and 11:00–17:00), with RD approaching 1.

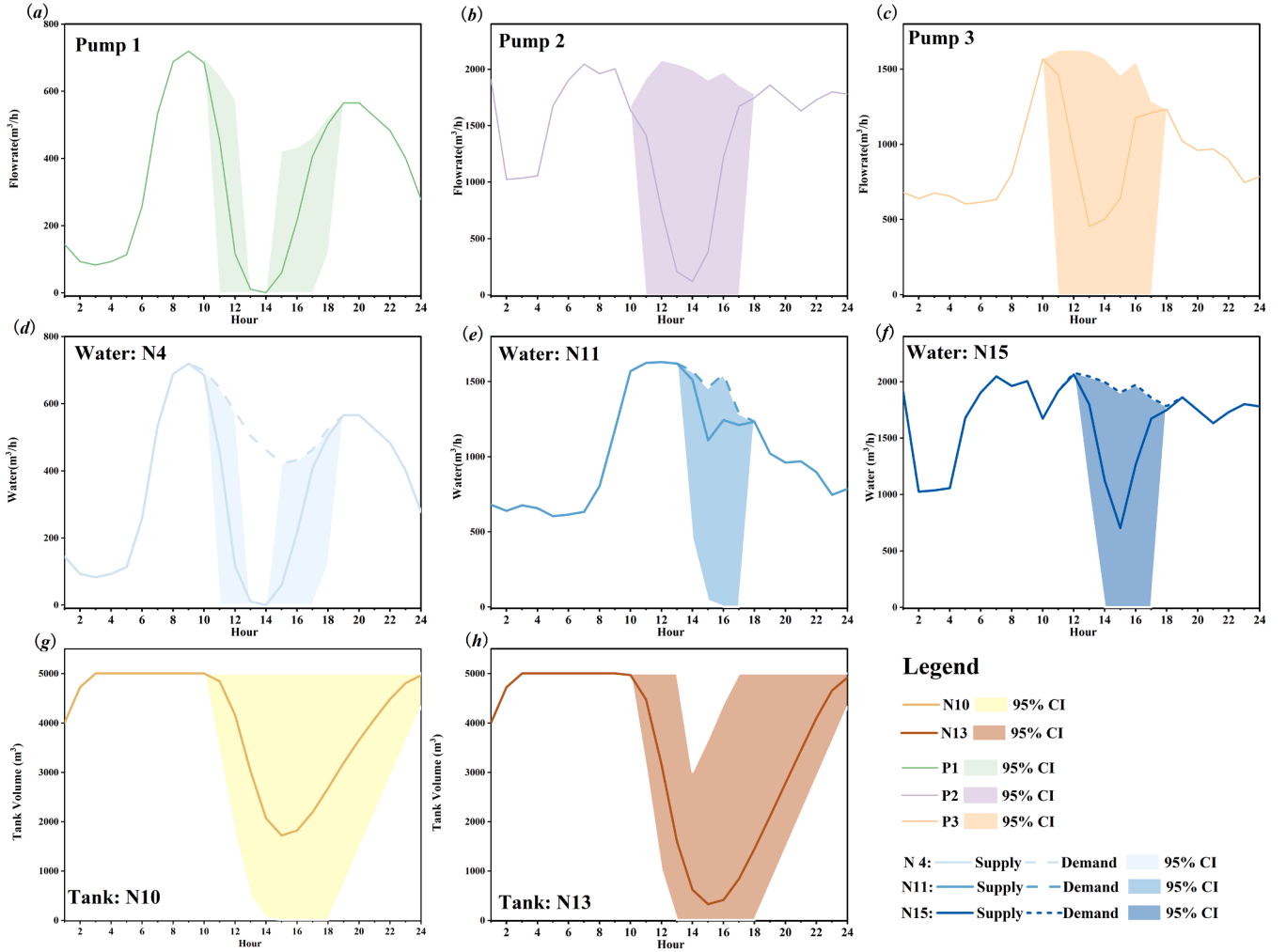


Fig. 8. Average flow evolution curves for P1 (a), P2 (b), and P3 (c); average actual water supply evolution curves for water demand nodes N4 (d), N11 (e), and N15 (f); average volume evolution curves for water storage tanks N10 (g) and N13 (h).

Similarly, in Scenario 16, the fault duration for P2 reached 6 h (12:00–17:00), with RD reaching 1. Conversely, RQ exhibits a significant positive correlation with P3 outage duration (0.53), yet shows no sensitivity to P1 and P2 outage durations. This indicates that the formation mechanism of RQ is highly dependent on P3. RA exhibits a monotonically consistent relationship with P3 duration (0.922, $p < 0.00001$). When RA exceeds the mean of 1.1, the average P3 duration exceeds 3 h.

5. Discussion

Unexpected or continuous power distribution system failures inevitably impact the normal operation of water distribution systems or other infrastructure being supported. Investigating the cascading impact mechanism of power outages on water distribution systems is critical to formulating strategies to enhance infrastructure resilience. However, there is a lack of studies that analyze the full process impacts of power interruptions on water distribution systems at the micro-scale, which makes it difficult to propose specific risk-prevention measures. This study assessed the vulnerability of water distribution systems by developing a dynamic cascade effect model of power outages on water systems and setting up random line failure scenarios. It aims to understand the interconnectivity of power and water distribution systems more accurately and to provide guidance for water management authorities in developing disaster response measures.

Compared with existing resilience and vulnerability assessment methods, the proposed indicators RD, RQ, and RA provide a process-oriented perspective for evaluating cascading impacts in water distribution systems. Previous studies proposed vulnerability indicators based on network connectivity, centrality, and others from a network topology perspective. For example, Shuang et al. [29] measure network vulnerability with network connectivity loss before and after failure. Tornyeviadzi et al. [32] combined the topological network indicator of closeness centrality and the hydraulic indicator of nodal water demand, constructing a comprehensive vulnerability index for cascading loss assessment. Although these metrics are useful for identifying structurally critical components, they neglect hydraulic operation and disturbance propagation processes. Cascading vulnerability assessments based on system performance evolution curves primarily focus on node failure probabilities [36] or evaluate socioeconomic losses from service disruptions, such as operational cost losses and socioeconomic vulnerability [42]. However, they inadequately account for components within water distribution systems, including pump flow rates, tank conditions, and node supply variations. In contrast, this study constructs the full-process evolution curves of SCP and SCW over the disaster lifecycle. Based on these curves, three metrics (RD, RQ, and RA) are developed to quantify performance degradation, shortage severity, and recovery characteristics. By capturing dynamic propagation and component-level operational responses, the proposed indicators provide a more physically meaningful basis for identifying critical elements and supporting

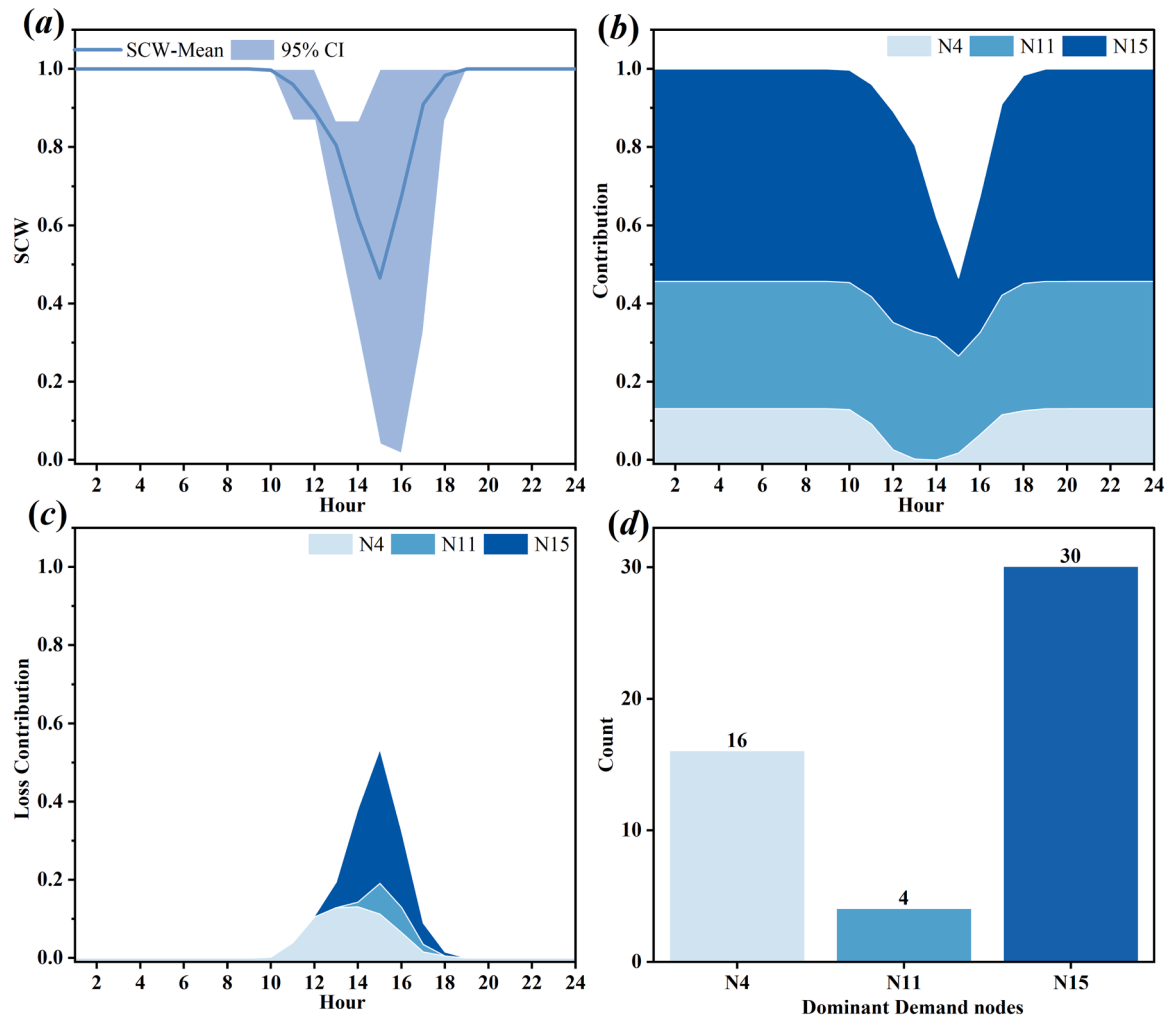


Fig. 9. Average SCW evolution curve under all scenarios (a); Contribution of actual water supply volume at each water demand node to SCW (b) and SCW loss (c); (d) Number of scenarios where each water demand node dominates SCW loss.

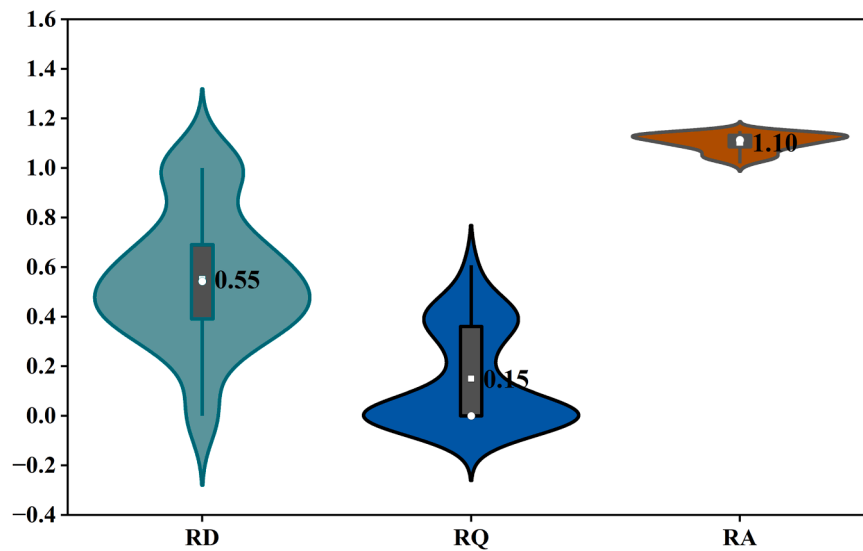


Fig. 10. Distribution of RD, RQ, and RA under all scenarios.

targeted resilience improvement in interdependent water–power systems.

The proposed method can be extended to larger or different types of water systems. For large-scale urban networks, the approach can be

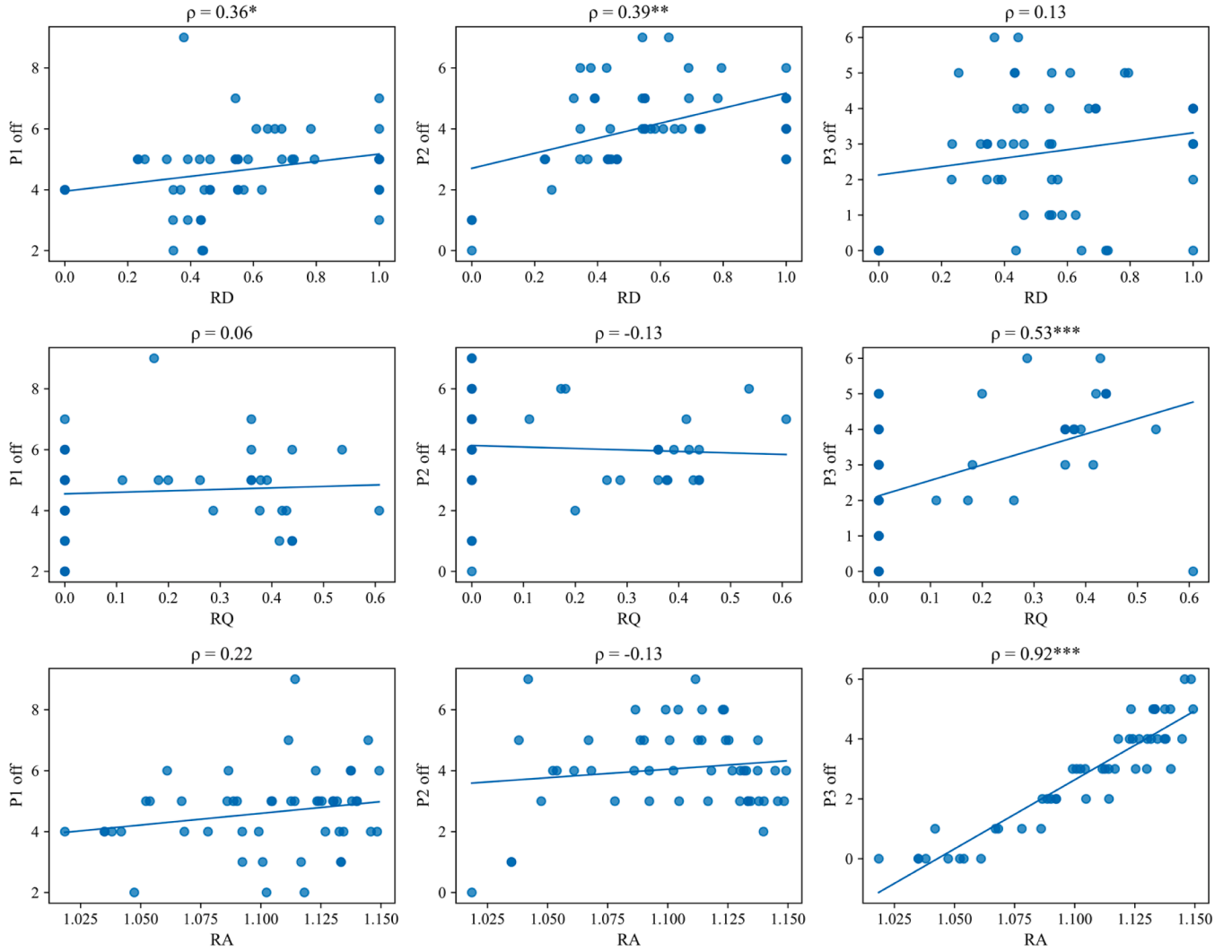


Fig. 11. Spearman correlation between pump outage hours and system performance indices (RD, RQ, and RA).

applied through network aggregation or sector-based decomposition, since the core indicators (SCP, SCW, RD, RQ, and RA) are computed from system-level performance curves rather than detailed node-by-node optimization. For rural or small-scale water infrastructures, the framework remains applicable by simplifying system components (e.g., limited pumps or storage units), while still capturing the coupling between power availability and water supply capacity. In addition, the disturbance generation module is scenario-driven and can incorporate extreme climate conditions (e.g., typhoons, heatwaves, prolonged outages) by modifying hazard intensity, duration, and repair-time distributions. Therefore, the proposed framework is adaptable to different system scales and hazard conditions.

This study has some limitations. Firstly, we focused only on the cascading impact of power interruptions on water supply losses, which is a one-way impact study. This methodology can be extended in the future to study the cascading impact of water supply interruptions on power generation losses, as well as the two-way impacts of cascading losses due to feedback loops. Then, additional system upgrade combinations can be considered, such as analyzing the effect of the installation location of distributed generation sources on the interdependence of water and power interconnection networks.

6. Conclusions

The study developed a cascading water shortage analysis model to

simulate how power interruptions affect water supply to demand nodes. Vulnerability metrics were also proposed to quantify cascading impacts on water and power interconnected systems. This methodology provides a framework for the analysis of system risk and resilience to aid in choosing design options or alternative operating strategies. The model was demonstrated on a case study using an IEEE 33-bus testing power distribution system and a 15-node water distribution system. Extreme storm scenarios were generated through Monte Carlo random sampling. Results show that during normal operation, P3 contributes most significantly to SCP, while in failure scenarios, P2 becomes the dominant factor in SCP losses, demonstrating its high vulnerability to power interruptions. During normal operation, N15 contributes most significantly to SCW, but both N15 and N4 contribute substantially to SCW losses. Therefore, increasing the backup water supply to N4 is key to enhancing the robustness of the water distribution system.

This study provides explicit guidelines for understanding cascading propagation in power and water distribution systems. Future research can consider how to improve water distribution system cascade vulnerability caused by power interruptions, which may require joint water and power optimization strategies, such as backup power sources, distribution line security reinforcement, and upgraded pumping stations, network design. Because of the global green energy transition, a similar approach can also be used to gauge the effects of diurnal or seasonal variability due to a high proportion of renewables in the grid.

Nomenclature

		Variables	
<i>Indices and Sets</i>		$q_i(t)$	Actual water supply to node i at time t [m^3]
$b, \mathcal{B}$	Index and set of power buses	$Q_{ji}(t)$	Pump flow rate from node j to node i [m^3]
$l, \mathcal{L}$	Index and set of power lines	$Q_{ik}(t)$	Pump flow rate from node i to node k [m^3]
$n, \mathcal{N}$	Index and set of water nodes	$\gamma_p(t)$	Failure state of pump p at time t [0,1]
$J, \mathcal{J}$	Index and set of water pipes	$\gamma_n(t)$	Supply rate of node n at time t [0,1]
t, T	Index and set for time	$Q_p(t)$	Water supply rate of pump p at time t [m^3]
p, P	Index and set for pumps	$P_p(t)$	Power of pump p at time t [kW]
$\mathcal{T}$	Set of water storage tanks	$H_p(t)$	Pump head [m]
i, j, k	Index of water demand nodes	$h_f(t)$	Friction loss in the pipeline at time t
<i>Parameters</i>		$H_{p,in}(t)$	Head at the pump inlet node at time t [m]
$D_i(t)$	water demand of node i at time t [m^3]	$H_{p,out}(t)$	Head at the pump outlet node at time t [m]
$Q_p^{\max}$	Maximum flow rate of pump p	$V_T(t)$	Water volume of tank T at time t [m^3]
ρ	Water density [1000 kg/m^3]	$Q_{T,t}^{\text{in}}$	Inflow rate of tank T at time t [m^3/h]
g	Gravitational acceleration [9.81 m/s^2]	$Q_{T,t}^{\text{out}}$	Outflow rate of tank T at time t [m^3/h]
η	Pump efficiency [0,1]	$SCW(t)$	Water service capacity at time t [0,1]
C	Roughness coefficient [0,1]	$SCP(t)$	Pump service capacity at Time t [0,1]
d	Pipe diameter [m]	$contrib_p(t)$	Contribution of pump p to SCP [0,1]
L	Pipe length [m]	$contrib_n(t)$	Contribution of node n to SCW [0,1]
a	Shut-off head	$Loss_p(t)$	Contribution of pump p to SCP loss [0,1]
b, c	Pump fitting parameters	$Loss_n(t)$	Contribution of node n to SCW loss [0,1]
$V_{T,\min}$	Minimum tank volume p [m^3]	$w_p(t)$	Contribution weight of pump p to SCP [0,1]
$V_{T,\max}$	Maximum tank volume [m^3]	$w_n(t)$	Contribution weight of node n to SCW [0,1]
$Q_{T,\max}$	Maximum charge/discharge rate of the water tank [m^3/h]	$t_{0,0}^p, t_{0,0}^w$	SCP, SCW degradation moment

(continued on next column)

(continued)

ε	Extremely small number	t_m^p, t_m^w	SCP, SCW lowest point moment
$\omega_{\max}$	Maximum wind speed	t_r^p, t_r^w	The time of SCP, SCW begins to restore
μ	Peak wind speed time	t_p^p, t_p^w	The time of SCP, SCW restored to the normal state
σ	The time scale of storm impacts	$A_{SCW}^{\text{deg}}, A_{SCP}^{\text{deg}}$	SCW, SCP degradation stage area
$\emptyset_l$	Line vulnerability coefficient	$A_{SCW}^{\text{rec}}, A_{SCP}^{\text{rec}}$	SCW, SCP degradation stage area
τ_l	Line repair time	$\omega(t)$	Wind speed at time t
μ_r	Average repair time	$x_l(t)$	Binary variable indicating line outage status
σ_r	Standard deviation of repair time	$p_l(t)$	Fault probability of the line at time t [0,1]

CRediT authorship contribution statement

Siqi Wang: Writing – review & editing, Methodology, Investigation, Formal analysis. **Fang Wang:** Writing – review & editing, Supervision, Project administration. **Yanmei Zhang:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Haoyuan Zhang:** Writing – review & editing, Visualization, Validation, Resources. **Zhiwei Li:** Writing – review & editing. **Tan Raymond:** Writing – review & editing. **Xiaoping Jia:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Appendix

Table A1
Power Distribution Lines Data

From bus	To bus	Resistance (Ohm)	Reactance (Ohm)
b1	b2	0.0922	0.047
b2	b3	0.493	0.2511
b3	b4	0.366	0.1864
b4	b5	0.3811	0.1941
b5	b6	0.819	0.707
b6	b7	0.1872	0.6188
b7	b8	1.7114	1.2351
b8	b9	1.03	0.74
b9	b10	1.044	0.74
b10	b11	0.1966	0.065
b11	b12	0.3744	0.1238
b12	b13	1.468	1.155
b13	b14	0.5416	0.7129
b14	b15	0.591	0.526
b15	b16	0.7463	0.545
b16	b17	1.289	1.721
b17	b18	0.732	0.574

(continued on next page)

Table A1 (continued)

From bus	To bus	Resistance (Ohm)	Reactance (Ohm)
b2	b19	0.164	0.1565
b19	b20	1.5042	1.3554
b20	b21	0.4095	0.4784
b21	b22	0.7089	0.9373
b3	b23	0.4512	0.3083
b23	b24	0.898	0.7091
b24	b25	0.896	0.7011
b6	b26	0.203	0.1034
b26	b27	0.2842	0.1447
b27	b28	1.059	0.9337
b28	b29	0.8042	0.7006
b29	b30	0.5075	0.2585
b30	b31	0.9744	0.963
b31	b32	0.3105	0.3619
b32	b33	0.341	0.5302

Table A2

Pump Performance Characteristic Function Data

Pumps	a	b	c	d	e
P1 (Arc 1)	178	95	2	295	134
P2 (Arc 4)	178	42	2	295	134
P3 (Arc 6)	178	30	2	295	134

Table A3

Water Node Data

Nodes	Elevation (m)	Minimum Pressure (m)	Maximum Pressure (m)
I1	10.6	0	1000
I2	9.7	20	1000
I3	0	20	1000
I4	5	20	1000
I5	55	20	1000
I6	126	20	1000
I7	121	20	1000
I8	126	20	1000
I9	157	20	1000
I10	156	20	1000
I11	156.65	20	1000
I12	128	20	1000
I13	124.13	20	1000
I14	123	20	1000
I15	106.42	20	1000

Table A4

Water Pipes Data

Pipes	Length (m)	Diameter (m)	Minimum FlowRate (m3/sec)	Maximum FlowRate (m3/sec)
J1	18	0.15	0	0.9
J2	401	0.15	0	0.9
J3	18	0.3	0	0.9
J4	1087	0.25	0	0.9
J5	1002	0.15	0	0.9
J6	846	0.1	0	0.9
J7	44	0.25	0	0.9
J8	18	0.25	0	0.9
J9	23	0.25	0	0.9
J10	16	0.1	0	0.9
J11	1003	0.2	0	0.9
J12	30	0.15	0	0.9
J13	6	0.23	0	0.9
J14	13	0.3	0	0.9

Table A5
Water Storage Tank Characteristic Data

Tanks	Volume (m ³)	Maximum Charge/Discharge Flow Rate (m ³ /hour)
T1 (Node 10)	5000	720
T2 (Node 13)	5000	720

Data availability

Data will be made available on request.

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